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United States Patent

12406783

Kind Code

B2

Date of Patent

September 02, 2025

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Circuit protection device

Abstract

A circuit protection device includes a first temperature sensitive resistor, a second temperature sensitive resistor, an electrically insulating multilayer, a first and second electrode layer, and at least one external electrode. The first temperature sensitive resistor and the second temperature sensitive resistor are electrically connected in parallel, and have a first upper electrically conductive layer and a second lower electrically conductive layer, respectively. The electrically insulating multilayer includes an upper insulating layer, a middle insulating layer, and a lower insulating layer. The upper insulating layer is between the first upper electrically conductive layer and the first electrode layer. The middle layer is laminated between the first temperature sensitive resistor and the second temperature sensitive resistor. The lower insulating layer is between the second lower electrically conductive layer and the second electrode layer. The external electrode is disposed on the first electrode layer, and extends beyond a peripheral wall along a horizontal direction.

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Appl. No.: 18/154599

Filed: January 13, 2023

Prior Publication Data

Document Identifier

Publication Date

US 20240071656 A1

Feb. 29, 2024

Foreign Application Priority Data

TW

111131804

Aug. 24, 2022

Publication Classification

Int. Cl.: H01C7/02 (20060101); H01C1/01 (20060101); H01C1/14 (20060101)

U.S. Cl.:

CPC H01C7/021 (20130101); H01C1/01 (20130101); H01C1/1406 (20130101);

Field of Classification Search

CPC: H01C (7/021); H01C (1/01); H01C (1/1406)

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Background/Summary

BACKGROUND OF THE INVENTION

(1) Field of the Invention

(1) The present application relates to a circuit protection device, and more specifically, to a thermally stable circuit protection device having temperature sensitive resistors electrically connected in parallel.

(2) Description of the Related Art

(2) Because the electrical resistance of conductive composite materials having a positive temperature coefficient (PTC) characteristic is very sensitive to temperature variation, it can be used as the material for current sensing devices and has been widely applied to over-current protection devices or circuit devices. More specifically, the electrical resistance of the PTC conductive composite material remains extremely low at normal temperatures, so that the circuit or cell can operate normally. However, when an over-current or an over-temperature event occurs in the circuit or cell, the electrical resistance will instantaneously increase to a high electrical resistance state (e.g., at least above $10^4 \Omega$), which is the so-called “trip.” Therefore, the over-current will be eliminated so as to protect the cell or the circuit device.

(3) Taking a polymeric PTC material as an example, carbon black is usually used as conductive filler which is evenly dispersed in crystalline polymer. In this crystalline structure, the carbon black particles are usually aligned at grain boundaries and are arranged closely. Accordingly, current can flow through the insulating crystalline polymer through such “carbon black chains.” At normal temperatures such as room temperature, numerous carbon chains exist in the polymer, constituting quite a lot of conductive paths. When the current passing through the conductive paths makes the temperature of device increase to a temperature exceeding the phase transition temperature of the polymer (e.g., the melting point of the polymer), the polymer would expand and these carbon black particles become separated from each other. As such, the carbon chains are broken and thus current is not allowed to pass therethrough, and as a consequence, the resistance increases instantaneously. The phenomenon of instant increase of resistance is the so-called “trip.”

(4) When the temperature decreases to below the phase transition temperature, the polymer is re-crystallized and the conductive carbon black chains are rebuilt. However, the polymer cannot be fully recovered after expansion so that the carbon chains cannot sustain original conductivity and the resistance cannot return to initial low resistance. After tripping many times, the resistance may increase significantly, resulting in poor resistance recovery or poor resistance repeatability.

(5) More specifically, when tripping, the volume of the PTC polymer changes tremendously, and the coefficient of thermal expansion (CTE) may be over 5000 ppm/K. Accordingly, the PTC material cannot fully return to its original structure owing to the tremendous change in volume, thereby significantly increasing the resistance when cooled back to room temperature. In a surface-mountable PTC device, the electrically conductive layers in physical contact with the PTC material layer are usually metal foils such as nickel foils, copper foils or nickel-plated copper foils. The CTE of the copper foil or nickel-plated copper foil are about 17 ppm/K, and the CTE of the nickel foil is 13 ppm/K, both are much smaller than that of the PTC polymer material. To solve this problem, traditionally, the manufacturers majorly concentrate improvement on structural design of the electrically conductive layers, by which the issue of poor resistance repeatability can be addressed. For example, one improvement is that the electrically conductive layers may be thickened so as to increase the structural strength of the PTC device.

(6) Moreover, it is understood that the development of electronic apparatuses grows fast as time goes on, and the electronic apparatuses to be protected require larger current flow more than ever for the purpose of quick charge and discharge. In this regard, the temperature sensitive resistor (i.e., two electrically conductive layers with one PTC material layer sandwiched therebetween) in a circuit protection device is often designed to have larger size, thereby decreasing the entire resistance as the surface area increases. However, the larger size of the circuit protection device means the larger space it occupies, and thus application of the circuit protection device may be limited as the electronic apparatuses become smaller and smaller. Even though the size of the circuit protection device could be reduced, it may encounter the issue of thermal stability. Heat generated by the circuit protection device easily accumulates, leading to a consequence that the circuit protection device has a low hold current ($I_{\text{sub.hold}}$). In some cases, the circuit protection device even expands and causes thermal stress after tripping, which may lead to the issue of poor

resistance recovery or poor resistance repeatability. Therefore, application of the circuit protection device may also be limited owing to the thermal stability described above.

(7) Apparently, miniaturization becomes the trend for the circuit protection device but it encounters lots of issues as described above. Accordingly, there is a need to improve the circuit protection device to solve the issues regarding resistance recovery, electrical resistance, and thermal stability.

SUMMARY OF THE INVENTION

(8) The present invention provides a circuit protection device with at least two temperature sensitive resistors. The temperature sensitive resistors are connected to each other in parallel, and therefore the circuit protection device may have lower electrical resistance so that the current flow is increased. In other words, the aforementioned parallel circuit allows the size of the circuit protection device to be reduced while maintaining low electrical resistance. However, electrical and structural characteristics of the circuit protection device are easily influenced by heat due to the reduction in size and/or the increase of current flow. As the issues of parallel temperature sensitive resistors described above, the present invention further adjusts the design of various components so as to improve the thermal stability of the circuit protection device. First, an upper insulating layer and a lower insulating layer are disposed below a top internal electrode (also called first electrode layer hereinafter) and above a bottom internal electrode (also called second electrode layer hereinafter), respectively. Both the upper insulating layer and the lower insulating layer have thermal expansion coefficients lower than that of a positive temperature coefficient (PTC) layer of the circuit protection device, by which structural strength of the circuit protection device is increased. In this way, the issue of poor resistance recovery incurred by thermal expansion can be solved. Second, at least one external electrode is provided. Its surface area, position to be placed, and length to be extended are taken into consideration and adjusted accordingly, by which heat dissipation is enhanced and hold current ($I_{sub,hold}$) of the circuit protection device may also be increased. Third, a plurality of notches are provided on one side of the circuit protection device, and stresses of thermal expansion are mitigated. Fourth, the circuit protection device is packaged with an insulation material, by which environment factors can be blocked and the entire structure of the circuit protection device may be further stabilized.

(9) In accordance with an aspect of the present invention, a circuit protection device has an upper surface, a lower surface opposite to the upper surface, and a peripheral wall connected between the upper surface and the lower surface. The circuit protection device includes a first temperature sensitive resistor, a second temperature sensitive resistor, an electrically insulating multilayer, a first electrode layer and a second electrode layer, and a first external electrode. The first temperature sensitive resistor includes a first upper electrically conductive layer, a first lower electrically conductive layer, and a first positive temperature coefficient (PTC) layer laminated between the first upper electrically conductive layer and the first lower electrically conductive layer. The second temperature sensitive resistor includes a second upper electrically conductive layer, a second lower electrically conductive layer, and a second PTC layer laminated between the second upper electrically conductive layer and the second lower electrically conductive layer. The electrically insulating multilayer has an upper insulating layer, a middle insulating layer, and a lower insulating layer. The upper insulating layer extends beyond the first upper electrically conductive layer, by which the first upper electrically conductive layer is entirely covered and the first PTC layer is partially covered by the upper insulating layer. The middle insulating layer is laminated between the first lower electrically conductive layer and the second upper electrically conductive layer, thereby bonding the first temperature sensitive resistor to the second temperature sensitive resistor. The lower insulating layer extends beyond the second lower electrically conductive layer, by which the second lower electrically conductive layer is entirely covered and the second PTC layer is partially covered by the lower insulating layer. The first electrode layer and the second electrode layer are attached to the upper insulating layer and the lower insulating layer, respectively. The first electrode layer is electrically connected to the first upper electrically

conductive layer and the second upper electrically conductive layer, and the second electrode layer is electrically connected to the first lower electrically conductive layer and the second lower electrically conductive layer. The first external electrode is disposed on the first electrode layer, and extends beyond the peripheral wall along a first horizontal direction parallel to the first electrode layer.

(10) In an embodiment, the circuit protection device further includes a right notch and a left notch. The upper insulating layer is attached between the first electrode layer and the first upper electrically conductive layer so as to space the first electrode layer apart from the first upper electrically conductive layer by a distance. The first electrode layer and the first upper electrically conductive layer are parallel to each other, and extend to the right notch, thereby electrically connecting to each other through the right notch.

(11) In an embodiment, the distance ranges from 0.02 mm to 0.06 mm.

(12) In an embodiment, an opening is formed on the first electrode layer, and is positioned directly above where the first PTC layer is partially covered by the upper insulating layer, by which the first electrode layer does not extend to the left notch and has no electrical connection to the left notch.

(13) In an embodiment, the circuit protection device further includes an electrically insulating material. The electrically insulating material is filled in the opening.

(14) In an embodiment, the right notch has a right electrically conductive member electrically connecting to the first electrode layer, the first upper electrically conductive layer, and the second upper electrically conductive layer. The left notch has a left electrically conductive member electrically connecting to the second electrode layer, the first lower electrically conductive layer, and the second lower electrically conductive layer.

(15) In an embodiment, the circuit protection device further includes a structural reinforcing metal film. The structural reinforcing metal film covers a surface of the first electrode layer, a surface of the second electrode layer, a surface of the right electrically conductive member, and a surface of the left electrically conductive member, by which the first electrode layer, the second electrode layer, the right electrically conductive member, and the left electrically conductive member are insulated from environmental air.

(16) In an embodiment, the peripheral wall has a left sidewall, a right sidewall opposite to the left sidewall, a front sidewall, and a rear sidewall opposite to the front sidewall. The left notch is disposed on the left sidewall, and the right notch is disposed on the right sidewall.

(17) In an embodiment, the first external electrode extends beyond the right sidewall along the first horizontal direction.

(18) In an embodiment, the circuit protection device has a first length parallel to the first horizontal direction, and the first external electrode has a second length from and beyond the right sidewall. If the sum of the first length and the second length is calculated as 100%, the second length accounts for 31% to 53%.

(19) In an embodiment, the second length ranges from 4 mm to 8 mm.

(20) In an embodiment, the upper surface of the circuit protection device has a first upper surface area, an upper surface of the first external electrode has a second upper surface area, and a value by dividing the second upper surface area by the first upper surface area ranges from 1.37 to 1.64.

(21) In an embodiment, the circuit protection device further includes a second external electrode disposed on the second electrode layer covered by the structural reinforcing metal film. The first external electrode extends along a second horizontal direction parallel to the first electrode layer and beyond the front sidewall. The second external electrode extends parallel to the first external electrode and beyond the rear sidewall.

(22) In an embodiment, the circuit protection device further includes a second external electrode disposed on the second electrode layer covered by the structural reinforcing metal film. The second external electrode extends opposite to the first horizontal direction and beyond the left sidewall.

(23) In an embodiment, the circuit protection device further includes a packaging tape. The

packaging tape wraps around the circuit protection device so as to cover the upper surface, the front sidewall, the lower surface, and the rear sidewall, by which only the right sidewall, the left sidewall, a part of the first external electrode, and a part of the second external electrode of the circuit protection device are exposed.

(24) In an embodiment, the circuit protection device further includes an electrically insulating frame. The electrically insulating frame encapsulates the circuit protection device, and only exposes a part of the first external electrode and a part of the second external electrode.

(25) In an embodiment, the circuit protection device further includes a plurality of right notches and a plurality of left notches opposite to the plurality of right notches. The plurality of right notches are disposed on a right sidewall and the plurality of left notches are disposed on a left sidewall of the peripheral wall, respectively, and the first external electrode extends beyond the right sidewall along the first horizontal direction.

(26) In an embodiment, the circuit protection device further includes a second external electrode disposed on the second electrode layer. The second external electrode extends opposite to the first horizontal direction parallel to the second electrode layer, and extends beyond the left sidewall.

(27) In an embodiment, the circuit protection device further includes a packaging tape. The packaging tape wraps around the circuit protection device, by which only the right sidewall, the left sidewall, a part of the first external electrode, and a part of the second external electrode of the circuit protection device are exposed.

(28) In an embodiment, the circuit protection device further includes an electrically insulating frame. The electrically insulating frame encapsulates the circuit protection device, and only exposes a part of the first external electrode and a part of the second external electrode.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present application will be described according to the appended drawings in which:

(2) FIG. 1 shows a three-dimensional view of a circuit protection device in accordance with a first aspect of the present invention;

(3) FIG. 2 shows an exploded view of the circuit protection device shown in FIG. 1;

(4) FIG. 3 shows a cross-sectional view of the circuit protection device shown in FIG. 1;

(5) FIG. 4 shows an enlarged view of a part of the circuit protection device shown in FIG. 3;

(6) FIG. 5 shows a top view of the circuit protection device shown in FIG. 1;

(7) FIG. 6 shows a top view of a circuit protection device in accordance with a second aspect of the present invention;

(8) FIG. 7 shows an exploded view of a circuit protection device in accordance with a third aspect of the present invention;

(9) FIG. 8 shows a flow chart for manufacturing the circuit protection device in accordance with the present invention;

(10) FIG. 9 shows a cross-sectional view of a circuit protection device in accordance with a fourth aspect of the present invention;

(11) FIG. 10 shows a three-dimensional view and an exploded view of a circuit protection device in accordance with a fourth aspect of the present invention; and

(12) FIG. 11 shows a three-dimensional view and an exploded view of a circuit protection device in accordance with a fifth aspect of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(13) The making and using of the presently preferred illustrative embodiments are discussed in detail below. It should be appreciated, however, that the present application provides many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The

specific illustrative embodiments discussed are merely illustrative of specific ways to make and use the invention, and do not limit the scope of the invention.

(14) Please refer to FIG. 1 to FIG. 5, in which a first aspect of the present invention is illustrated in three-dimensional view, cross-sectional view, enlarged view, and top view. In FIG. 1, a circuit protection device **100** has a PTC chip as its core structure. The PTC chip has a left notch **62** and a right notch **61**. A first external electrode **51** is disposed on an upper surface **101** of the circuit protection device **100**, and a second external electrode **52** is disposed on a lower surface **102** of the circuit protection device **100**. For ease of understanding and discussion, the first external electrode **51** and the second external electrode **52** are detached from the PTC chip in FIG. 2. In addition, x-axis, y-axis, and z-axis in the figure show orientation in space, and therefore the present invention can be fully discussed in different views. The circuit protection device **100** has an upper surface **101**, a lower surface **102** opposite to the upper surface **101**, and a peripheral wall **103** connected between the upper surface **101** and the lower surface **102**. The peripheral wall **103** has a left sidewall **S2**, a right sidewall **S1**, a front sidewall **S4**, and a rear sidewall **S3**. The left sidewall **S2** is opposite to the right sidewall **S1**, and the front sidewall **S4** is opposite to the rear sidewall **S3**. The left notch **62** is disposed on the left sidewall **S2**, and the right notch **61** is disposed on the right sidewall **S1**. The upper surface **101**, the lower surface **102**, the left notch **62**, and the right notch **61** are covered by a structural reinforcing metal film **80** in order to increase structural strength while preventing the electrode layer (first electrode layer **41** and second electrode layer **42**) and conductive member (right electrically conductive member **63** and left electrically conductive member **64**) from oxidation. The structural reinforcing metal film **80** may be tin foil, nickel foil, nickel plated copper foil, or the like, which can be formed by sputtering or electroplating.

(15) Please refer to FIG. 3, a cross-sectional view is obtained by taking along the dashed line AA in FIG. 1. The circuit protection device **100** includes a first temperature sensitive resistor **10**, a second temperature sensitive resistor **20**, an electrically insulating multilayer **30**, a first electrode layer **41**, a second electrode layer **42**, the left notch **62**, the right notch **61**, and at least one external electrode (e.g., first external electrode **51** and/or second external electrode **52**). The first electrode layer **41** is electrically connected to the first temperature sensitive resistor **10** and the second temperature sensitive resistor **20** through the right notch **61**. The second electrode layer **42** is electrically connected to the first temperature sensitive resistor **10** and the second temperature sensitive resistor **20** through the left notch **62**. The first electrode layer **41** and the second electrode layer **42** may be referred to as internal electrodes of the circuit protection device **100**. In an embodiment, the circuit protection device **100** may not include the external electrodes **51**, **52** and may be manufactured as surface-mountable type device (SMD) that can be directly welded to circuit board. The first temperature sensitive resistor **10** includes a first upper electrically conductive layer **12**, a first lower electrically conductive layer **13**, and a first positive temperature coefficient (PTC) layer **11** laminated between the first upper electrically conductive layer **12** and the first lower electrically conductive layer **13**. The second temperature sensitive resistor **20** includes a second upper electrically conductive layer **22**, a second lower electrically conductive layer **23**, and a second PTC layer **21** laminated between the second upper electrically conductive layer **22** and the second lower electrically conductive layer **23**. Moreover, the circuit protection device **100** has a first length **L1**, and the external electrode (e.g., the first external electrode **51**) has a second length **L2** extending from and beyond the peripheral wall **103**, and the total length of the external electrode is referred to as a third length **L3**. It is noted that the circuit protection device **100** includes at least two temperature sensitive resistors (i.e., the first temperature sensitive resistor **10** and the second temperature sensitive resistor **20**) having PTC layers therein, and the temperature sensitive resistors are connected to each other in parallel. Compared with the conventional circuit protection device of the same size and having only one temperature sensitive resistor, the circuit protection device **100** significantly increases the surface area of the electrode layer and correspondingly decreases the electrical resistance owing to the parallel connection between the temperature sensitive resistors. In

other words, the circuit protection device **100** may have lower electrical resistance so that the current flow is increased. In addition, to further enhance the thermal stability of the parallel configuration, the present invention adjusts the design of various components, as detailed below.

(16) Please refer to FIG. 3 and FIG. 4. The electrically insulating multilayer **30** has an upper insulating layer **31**, a middle insulating layer **32**, and a lower insulating layer **33**. The upper insulating layer **31** extends beyond the first upper electrically conductive layer **12**, by which the first upper electrically conductive layer **12** is entirely covered, and the first PTC layer **11** is partially covered. The middle insulating layer **32** is laminated between the first lower electrically conductive layer **13** and the second upper electrically conductive layer **22**, thereby bonding the first temperature sensitive resistor **10** to the second temperature sensitive resistor **20**. The lower insulating layer **33** extends beyond the second lower electrically conductive layer **23**, by which the second lower electrically conductive layer **23** is entirely covered, and the second PTC layer **21** is partially covered. In an embodiment, the electrically insulating multilayer **30** may be a material selected from the group consisting of glass fiber reinforced epoxy resin composite, epoxy resin, polyester resin, polyamide resin, phenolic resin, polyurethane resin, polycyanate resin, maleimide resin, polyolefin resin, polyphenylene oxide resin, photopolymer resin, and combinations thereof. The coefficient of thermal expansion (CTE) of the material is lower than that of the PTC layers **11**, **21**. In addition, the middle insulating layer **32** may further mix inorganic functional micro- or nanoparticles in the material for the purpose of improvement of adhesive property, structural strength, or other physical properties.

(17) More specifically, the upper insulating layer **31** is attached between and is physically in contact with the first electrode layer **41** and the first upper electrically conductive layer **12**, so as to space the first electrode layer **41** apart from the first upper electrically conductive layer **12** by a distance D . The first electrode layer **41** and the first upper electrically conductive layer **12** are parallel to each other and extend along y-axis to the right notch **61**, electrically connecting to each other through the right notch **61**. In addition, an opening O is formed on the first electrode layer **41**, and is positioned directly above (along z-axis) where the first PTC layer **11** is partially covered by the upper insulating layer **31**, by which the first electrode layer **41** does not extend to the left notch **62** and has no electrical connection to the left notch **62**. Moreover, there is not any metal layer extending from the left notch **62** to cover the upper insulating layer **31** on the horizontal plane (i.e., x-y plane) where the first electrode layer **41** is located. In other words, except for the first electrode layer **41**, there is no other electrode layer existing on the same plane as the first electrode layer **41** at the other side of the circuit protection device **100** to cover the upper insulating layer **31**.

Therefore, there would be no likelihood that the first external electrode **51** may possibly contact the above-said other electrode layer and the possibility of short circuit is reduced even if the welding position of the first external electrode **51** shifts on x-y plane owing to alignment error. In an embodiment, the circuit protection device **100** further includes an electrically insulating material **70**, such as solder mask or solder resist. The electrically insulating material **70** is filled in the opening O , so as to increase electrical insulation between the first electrode layer **41** and the left notch **62**, and avoid improper welding because of poor weldability of the electrically insulating material **70**. The lower insulating layer **33** is attached between and is physically in contact with the second electrode layer **42** and the second lower electrically conductive layer **23**. Another opening O is also formed on the second electrode layer **42**, and is positioned directly below where the first PTC layer **11** is partially covered by the lower insulating layer **33**. The configuration of the opening O below the lower insulating layer **33** is the same as that of the above-said upper insulating layer **31**, and is not described in detail herein.

(18) In the present disclosure, the CTE of the upper insulating layer **31** is far different from that of the first PTC layer **11**, that is, the CTE of the upper insulating layer **31** is lower than that of the first PTC layer **11**. Therefore, the upper insulating layer **31** can suppress the first PTC layer **11** from excessive expansion, and increase the structural strength of the circuit protection device **100**, by

which the issue of poor resistance recovery incurred by thermal expansion can be solved. In addition, in this embodiment, the electrically conductive layer is divided into two parts (i.e., first upper electrically conductive layer **12** and first electrode layer **41**), both connecting to the right notch **61** on the same side, thereby increasing the surface area of the conductor in a single temperature sensitive resistor **10**. More specifically, because the first electrode layer **41** is separated from the first upper electrically conductive layer **12** by the upper insulating layer **31**, the bottom surface of the first electrode layer **41** is not coplanar with the top surface of the first upper electrically conductive layer **12**. As a result, this increases the surface area of the conductor and in the meanwhile, lowers electrical resistance of the temperature sensitive resistor **10**. The lower insulating layer **33**, the second lower electrically conductive layer **23**, and the second electrode layer **42** are symmetric with and may have the same configuration as the upper insulating layer **31**, the first upper electrically conductive layer **12**, and the first electrode layer **41**. Since the upper insulating layer **31** and the lower insulating layer **33** clamp two temperature sensitive resistors (i.e., first temperature sensitive resistor **10** and second temperature sensitive resistor **20**) therebetween from top and bottom, the thermal expansion of the circuit protection device **100** can be suppressed, and the circuit protection device **100** can be prevented from deformation. Particularly, both the upper insulating layer **31** and the lower insulating layer **33** do not have a thick thickness, both of which would not occupy much space in the circuit protection device **100** and therefore the space occupation rate is low. For example, the distance D (see FIG. 4) may range from about 0.02 mm to 0.06 mm. In an embodiment, the distance D may be 0.02 mm, 0.03 mm, 0.04 mm, 0.05 mm, or 0.06 mm. In addition, the first electrode layer **41** has a fourth length L4, and the first upper electrically conductive layer **12** has a fifth length L5 in the direction along y-axis. The fourth length L4 and the fifth length L5 are longer than a half of total length of the circuit protection device **100** along y-axis, facilitating in that the circuit protection device **100** has a desired low electrical resistance. Generally, considering there is a concern that the metal surface (e.g., the upper surface) of the first electrode layer **41** may have too large area exposed to the ambient environment, size of the first electrode layer **41** is preferably smaller than that of the first upper electrically conductive layer **12**. That is, it is preferred that the fourth length L4 is shorter than the fifth length L5. In an embodiment, the first length L1 ranges from about 7 mm to 9 mm, such as 7 mm, 7.3 mm, 7.9 mm, 8.2 mm, 8.7 mm, or 9 mm. Alternatively, the fourth length L4 may be longer than the fifth length L5. However, the fourth length L4 should be tightly controlled and the first electrode layer **41** would not be disposed so close to the left notch **62** that electrical arcing or even short-circuit might occur. In short, the upper insulating layer **31** and the lower insulating layer **33** occupy little space, and existence thereof may increase the surface area of the conductor and the structural strength of the circuit protection device **100**.

(19) Please refer to FIG. 3, the first electrode layer **41** and the second electrode layer **42** are attached to the upper insulating layer **31** and the lower insulating layer **33**, respectively. The first electrode layer **41** is electrically connected to the first upper electrically conductive layer **12** and the second upper electrically conductive layer **22**, and the second electrode layer **42** is electrically connected to the first lower electrically conductive layer **13** and the second lower electrically conductive layer **23**. More specifically, the right notch **61** has a right electrically conductive member **63** electrically connecting to the first electrode layer **41**, the first upper electrically conductive layer **12**, and the second upper electrically conductive layer **22**. The first electrode layer **41**, the first upper electrically conductive layer **12**, and the second upper electrically conductive layer **22** are electrically connected to each other through the right electrically conductive member **63**. The left notch **62** has a left electrically conductive member **64** electrically connecting to the second electrode layer **42**, the first lower electrically conductive layer **13**, and the second lower electrically conductive layer **23**. The second electrode layer **42**, the first lower electrically conductive layer **13**, and the second lower electrically conductive layer **23** are electrically connected to each other through the left electrically conductive member **64**. It is noted that, as

described above, the circuit protection device **100** may further include the structural reinforcing metal film **80**. The structural reinforcing metal film **80** covers a surface of the first electrode layer **41**, a surface of the second electrode layer **42**, a surface of the right electrically conductive member **63**, and a surface of the left electrically conductive member **64**, by which the first electrode layer **41**, the second electrode layer **42**, the right electrically conductive member **63**, and the left electrically conductive member **64** are insulated from environmental air. The coefficient of thermal expansion (CTE) of the structural reinforcing metal film **80** is much lower than that of the PTC layers **11**, **21**, and therefore the structural strength of the circuit protection device **100** is enhanced. In addition, the structural reinforcing metal film **80** is less prone to oxidation and has excellent conductivity, and it may cover the exposed portion of metal electrodes (i.e., first electrode layer **41** and second electrode layer **42**) which are easily oxidized. Please refer to FIG. 2, the structural reinforcing metal film **80** covers the first electrode layer **41** so that a part of the upper surface **101** of the circuit protection device **100** is covered by the structural reinforcing metal film **80**, and the remaining part of the upper surface **101** is covered by the electrically insulating material **70**. On the right sidewall **S1**, the overall exposed surface of the right electrically conductive member **63** in the right notch **61** is covered by the structural reinforcing metal film **80**. In addition, the structural reinforcing metal film **80** may have the same material as the solder, which is favorable to weldability of different components to be joined. For example, the structural reinforcing metal film **80** may be a metal film including tin. If the solder consists of tin alloy, weldability between the solder and the structural reinforcing metal film **80** is increased. The lower surface **102** and the left notch **62** may have the same design as the upper surface **101** and the right notch **61**, and are not described in detail herein.

(20) Please refer to FIG. 5 and FIG. 2, in which the circuit protection device **100** is illustrated in top view. For ease of understanding, the discussion below mainly focuses on the first external electrode **51**, and it is noted that the second external electrode **52** may have the same design as the first external electrode **51**. The first external electrode **51** is disposed on the first electrode layer **41**, and extends beyond the peripheral wall **103** along a first horizontal direction parallel to the first electrode layer **41**. More specifically, the first external electrode **51** may extend in the first horizontal direction along y-axis and away from the left sidewall **S2**, and then extend beyond the right sidewall **S1**. Therefore, the right sidewall **S1** and the right notch **61** thereon, illustrated as the dashed line shown in FIG. 5, are covered by the first external electrode **51**. The first horizontal direction is substantially parallel to y-axis. The circuit protection device **100** has the first length **L1** parallel to the first horizontal direction, and the first external electrode **51** has the second length **L2** from and beyond the right sidewall **S1** (also as shown in FIG. 2 and FIG. 3). If the sum of the first length **L1** and the second length **L2** is calculated as 100%, the second length **L2** accounts for 31% to 53%. The second length **L2** may be referred as “extending ratio” of the external electrode. In an embodiment, the extending ratio may be 31%, 33%, 36%, 38%, 42%, 46%, 47%, 49%, or 53%. For example, if the first length **L1** is 8.2 mm and the second length **L2** is 5 mm, the extending ratio would be calculated as about $(5/(8.2+5))*100\%=38\%$. In this way, heat dissipation is enhanced and hold current ($I_{sub,hold}$) of the circuit protection device **100** may also be increased due to additional extension of the length of external electrode that increases the surface area of the first external electrode **51** to be exposed to the environment. The improvement may be applied to different sizes of the circuit protection device, as long as the extension of the second length **L2** is adjusted in a suitable range. If the extending ratio is lower than 31%, the first external electrode **51** is too short to be welded to an external device. If the extending ratio is higher than 53%, the first external electrode **51** is easily bent and prone to deformation due to its excessive length. Also, entire size of the circuit protection device **100** is unnecessarily increased and such larger device occupies much space if the extending ratio is higher than 53%. In a preferred embodiment, the extending ratio is higher than 47%, such as in a range from 47% to 53%, in order to increase the quality of welding (e.g., spot welding) and heat dissipation. As for the material of the external

electrode, the first external electrode **51** and the second external electrode **52** may be copper strips, nickel strips, or steel strips. Shape of the nickel strip is easily influenced by heat, and thus the nickel strip is suitable for spot welding rather than reflow welding. Structural strength of the copper strip is better than that of the nickel strip, and thus the copper strip is suitable for reflow welding.

(21) In FIG. 5, the circuit protection device **100** further has a first width **W1**, and each of the first external electrode **51** and the second external electrode **52** has a second width **W2**. In an embodiment, the first width **W1** ranges from 4 mm to 6 mm, such as 4 mm, 4.3 mm, 4.5 mm, 5.3 mm, 5.7 mm, or 6 mm. For ease of assembly, the second width **W2** is preferably equal to or narrower than the first width **W**. The extension length of the first external electrode **51** (i.e., the second length **L2**) is helpful in heat dissipation as described above. In addition, the surface area on top of the first external electrode **51** can also be adjusted. More specifically, the first length **L1** times the first width **W1** is equal to the surface area of the upper surface **101** of the circuit protection device **100** (referred to as “first upper surface area” hereinafter), and the third length **L3** (as shown in FIG. 3) times the second width **W2** is equal to the surface area of the upper surface of the first external electrode **51** (referred to as “second upper surface area” hereinafter). In an embodiment, the third length **L3** ranges from about 9.25 mm to 12.25 mm, such as 9.25 mm, 10.25 mm, 11.25 mm, or 12.25 mm. For further improvement in heat dissipation, the second upper surface area of the first external electrode **51** is adjusted, and a value by dividing the second upper surface area by the first upper surface area ranges from 1.37 to 1.64. For example, the circuit protection device **100** has the first length **L1** with 8.2 mm, and the first width **W1** with 5.3 mm; and for further improvement in heat dissipation, the third length **L3** can be set up to 12.25 mm, and the second width **W2** is 5.3 mm which is equal to the first width **W1**. Accordingly, the second upper surface area and the first upper surface area can be calculated, and are about 64.93 mm² and 43.46 mm², respectively, and the ratio therebetween is about 1.49. In this way, the second upper surface area of the first external electrode **51** is larger than the first upper surface area of the circuit protection device **100**, and the ratio therebetween is in the range from 1.37 to 1.64, by which the first external electrode **51** is helpful in heat dissipation and hold current ($I_{sub,hold}$) of the circuit protection device **100** is increased. Moreover, in another embodiment, the first external electrode **51** having the second upper surface area of 64.93 mm² can be installed away from the right sidewall **S1**, which makes the second length **L2** equal to 8 mm and has the best extending ratio. In other words, the present disclosure not only adjusts the second upper surface area of the first external electrode **51**, but also installs the first external electrode **51** in the direction away from the peripheral wall **103**. Even if the first external electrode **51** is covered by an insulation material in the subsequent packaging process, a portion of the upper surface of the first external electrode **51** still remains uncovered and exposed to the environment, thereby effectively enhancing heat dissipation and increasing hold current ($I_{sub,hold}$) of the circuit protection device **100**. In an embodiment, there are circuit protection devices **100** in different sizes, and hence the ratio between the second upper surface area and the first upper surface area may be 1.37, 1.43, 1.49, 1.55, 1.58, 1.61, 1.64. In a preferred embodiment, the value by dividing the second upper surface area by the first upper surface area ranges from 1.49 to 1.64, and the circuit protection device **100** has larger hold current ($I_{sub,hold}$).

(22) Please refer to FIG. 6, in which a circuit protection device **200** is illustrated in accordance with a second aspect of the present invention. Compared with FIG. 5, extension direction of the first external electrode **51** and the second external electrode **52** in FIG. 6 is different. In FIG. 5, the first external electrode **51** of the circuit protection device **100** extends beyond the right sidewall **S1** along the first horizontal direction, and is disposed on the first electrode layer **41** covered by the structural reinforcing metal film **80**. The second external electrode **52** extends opposite to the first horizontal direction and beyond the left sidewall **S2**, and is disposed on the second electrode layer **42** covered by the structural reinforcing metal film **80**. In contrast, in FIG. 6, the first external electrode **51** of the circuit protection device **200** extends along a second horizontal direction

parallel to the first electrode layer **41** and beyond the front sidewall **S4**. The second external electrode **52** extends parallel to the first external electrode **51** and beyond the rear sidewall **S3**. That is, the first external electrode **51** and the second external electrode **52** extend outward, and are perpendicular to the long side of the circuit protection device **200**. The first horizontal direction is substantially parallel to y-axis, and the second horizontal direction is substantially parallel to x-axis; and hence on x-y plane, the first horizontal direction is substantially perpendicular to the second horizontal direction. Accordingly, the first external electrode **51** and the second external electrode **52** may vary in configuration depending on the requirements.

(23) Please refer to FIG. 7, in which a circuit protection device **300** is illustrated in accordance with a third aspect of the present invention. In FIG. 7, the same features as shown in the first aspect are indicated by the same numerals, and are not described in detail herein. Compared with FIG. 2, FIG. 7 shows a circuit protection device **300** having different number and position of notches. More specifically, the circuit protection device **300** further includes a plurality of right notches (e.g., a first right notch **65** and a second right notch **66**) and a plurality of left notches (e.g., a first left notch **67** and a second left notch **68**) opposite to the plurality of right notches. The plurality of right notches are disposed on the right sidewall **S1** and the plurality of left notches are disposed on the left sidewall **S2** of the peripheral wall **103**, respectively. Adding number of notches on one side can mitigate stresses caused by thermal expansion, and therefore, no crack would be produced on the circuit protection device **300** during use of device. Furthermore, the mitigation and distribution of stresses would be more obvious if the notches are symmetric on both sides (i.e., the right sidewall **S1** to the left sidewall **S2**). In an embodiment, the first right notch **65** and the second right notch **66** are recesses in a shape of half-cylinder, and both of them may have the same radius in top view. In another embodiment, the present disclosure designs the notches in a more symmetric manner, and therefore the first right notch **65** and the second right notch **66** are symmetric on the right sidewall **S1**. That is, the right sidewall **S1** is equally divided into three rectangular sidewalls, all of which have the same surface area. The same configuration may be applied to the left sidewall **S2**. Moreover, the first external electrode **51** may extend beyond the right sidewall **S1** along the first horizontal direction, and is disposed on the first electrode layer **41** covered by the structural reinforcing metal film **80**. The second external electrode **52** may extend opposite to the first horizontal direction and beyond the left sidewall **S2**, and is disposed on the second electrode layer **42** covered by the structural reinforcing metal film **80**. In another embodiment, the first external electrode **51** may extend along the second horizontal direction and beyond the front sidewall **S4**, and the second external electrode **52** may extend parallel to the first external electrode **51** and beyond the rear sidewall **S3**. In other words, orientation between the first external electrode **51** and the second external electrode **52** may be the same as that in either FIG. 5 or FIG. 6.

(24) Please further refer to FIG. 8, in which a manufacturing process of the circuit protection device **100** (shown in FIG. 1) is illustrated for ease of understanding the present disclosure. The manufacturing process generally includes step 1 to step 7. Details are described below.

(25) Step 1: A first upper electrically conductive plate **120**, a first lower electrically conductive plate **130**, and a first positive temperature coefficient plate **110** are provided. The first positive temperature coefficient plate **110** is laminated between the first upper electrically conductive plate **120** and the first lower electrically conductive plate **130** by performing a hot-pressing process. Then, the first upper electrically conductive plate **120** and the first lower electrically conductive plate **130** are etched to expose the first positive temperature coefficient plate **110**, and to form a plurality of first upper electrically conductive layers **12** and a plurality of first lower electrically conductive layers **13**.

(26) Step 2: The step 1 is repeated again, and a second positive temperature coefficient plate **210**, a plurality of second upper electrically conductive layers **22**, and a plurality of second lower electrically conductive layers **23** are obtained. Next, a first electrode plate **410** and an upper insulating plate **310** are sequentially stacked on the plurality of first upper electrically conductive

layers **12**; a middle insulating plate **320** is positioned between the plurality of first lower electrically conductive layers **13** and the plurality of first upper electrically conductive layers **12**; and a lower insulating plate **330** and a second electrode plate **420** are sequentially stacked below the plurality of second lower electrically conductive layers **23**.

(27) Step 3: The aforementioned layers are stacked in an order like that in the step 2, and are hot-pressed accordingly. Therefore, a laminated structure having two positive temperature coefficient plates is obtained.

(28) Step 4: The laminated structure in the step 3 is drilled to form a plurality of electrically conductive vias H. The drilling method may be mechanical drilling, laser drilling, or other drilling techniques. After drilling, a plating process is performed on the plurality of electrically conductive vias H. The plurality of electrically conductive vias H is not limited to plating through holes (PTH). As long as electrical connection can be made, PTH could be replaced with electrical pins or other similar structures instead. Accordingly, a plurality of right electrically conductive members **63** and a plurality of left electrically conductive members **64** are formed in the plurality of electrically conductive vias H. In the meantime, a plurality of upper insulating layers **31**, a plurality of first positive temperature coefficient layers **11**, a plurality of middle insulating layers **32**, a plurality of first positive temperature coefficient layers **21**, and a plurality of lower insulating layers **33** are formed between the plurality of electrically conductive vias H.

(29) Step 5: After drilling in the step 4, the first electrode plate **410** and the second electrode plate **420** are etched to form a plurality of openings O, thereby exposing the plurality of upper insulating layers **31** and the plurality of lower insulating layers **3**, and forming a plurality of first electrode layers **41** and a plurality of second electrode layers **42** thereon, respectively.

(30) Step 6: In order to increase structural strength of the circuit protection device **100** and prevent the metal electrodes from oxidation, surfaces of the plurality of first electrode layers **41**, the plurality of second electrode layers **42**, the plurality of right electrically conductive members **63**, and the plurality of left electrically conductive members **64** are covered by a structural reinforcing metal film **80**. In an embodiment, the structural reinforcing metal film **80** may be a tin-containing film, and surfaces of the plurality of first electrode layers **41**, the plurality of second electrode layers **42**, the plurality of right electrically conductive members **63**, and the plurality of left electrically conductive members **64** may be coated by the tin-containing film through a sputtering or an electroplating process.

(31) Step 7: The structure in the step 6 is diced along a dicing line d1-d1, a dicing line d2-d2, a dicing line d3-d3, and a dicing line d4-d4. Therefore, at least three circuit protection devices are produced as shown in the figure.

(32) In addition to the foregoing steps, a packaging process may be further performed for the purpose of enhancement of insulation and structural stability of the device. Therefore, the circuit protection device may have a packaging structure as shown in FIG. 9 to FIG. 11.

(33) FIG. 9 and FIG. 10 show a circuit protection device **400** in accordance with a fourth aspect of the present invention. In FIG. 9, the circuit protection device **400** further includes a packaging tape **90**. The packaging tape **90** may be a film-like tape made of polymers and has adhesive property. For example, the packaging tape **90** may be made from a polymer matrix and an adhesive. The polymer matrix may be polyester film, and the adhesive may be silicon gel. In FIG. 10, it is shown that the packaging tape **90** wraps around and is adhered to the circuit protection device **400** so as to cover the upper surface **101**, the front sidewall S4, the lower surface **102**, and the rear sidewall S3, by which only the right sidewall S1, the left sidewall S2, a part of the first external electrode **51**, and a part of the second external electrode **52** of the circuit protection device **400** are exposed. It is noted that, for ease of understanding, the upper figure in FIG. 10 shows a partially exploded view of the circuit protection device **400**, in which the first external electrode **51** and the second external electrode **52** are detached from the circuit protection device **400**. As can be seen, the packaging tape **90** does not cover the right sidewall S1 and the left sidewall S2. The packaging tape

90 has two sides which may extend beyond the right sidewall **S1** and the left sidewall **S2** along y-axis, or may be flush with the right sidewall **S1** and the left sidewall **S2**. According to the present invention, in case that right and left sides of the packaging tape **90** extend beyond the peripheral wall **103**, the second length **L2** is calculated from the packaging tape **90** along y-axis, as shown in FIG. 9. Because the circuit protection device **400** is enclosed by the packaging tape **90**, the circuit protection device **400** may have better insulation and structural stability. Moreover, the packaging tape **90** may be either transparent or non-transparent. In the case that the packaging tape **90** is transparent, such a transparent packaging tape **90** would be helpful in defect inspection. Any person may employ his naked eyes or an optical device could be used to check whether there are defects existing in the circuit protection device **400** even though covered by the packaging tape **90**. In this way, transparency of the packaging tape **90** is favorable to quality control and increase of yield rate of production. It is understood that features of the fourth aspect (except the packaging tape **90**) may be the same as that of the first aspect, the second aspect, and the third aspect, and are not described in detail herein.

(34) FIG. 11 shows a circuit protection device **500** in accordance with a fifth aspect of the present invention. Compared with FIG. 10, the circuit protection device **500** in FIG. 11 is packaged by an electrically insulating frame **91** rather than the packaging tape **90**. The electrically insulating frame **91** may be a cured thermosetting polymer. More specifically, the circuit protection device **500** may be coated by a flowable thermosetting polymer through injection molding or other packaging processes, and then the flowable thermosetting polymer is cured thereon. Therefore, the electrically insulating frame **91** encapsulates the circuit protection device **500**, and only exposes a part of the first external electrode **51** and a part of the second external electrode **52**. It is understood that features of the fifth aspect (except the electrically insulating frame **91**) may be the same as that of the first aspect, the second aspect, the third aspect, and the fourth aspect, and are not described in detail herein.

(35) To sum up, according to the present invention, a structural reinforcing metal film **80** is coated on the surface of the circuit protection device **100**, and the circuit protection device **100** includes an upper insulating layer **31** and a lower insulating layer **33**, therefore structural strength of the device can be increased, and the issue of poor resistance recovery incurred by thermal expansion can be solved. Moreover, the present invention also makes changes for the external electrodes **51**, **52**, including their surface area, their position to be placed, and their length to be extended, therefore heat dissipation efficiency of the circuit protection device can be enhanced and hold current (I.sub.hold) of the circuit protection device is also increased. Accordingly, the circuit protection device can be applied to electronic apparatuses which need higher rated current. In addition, the present invention forms at least one right notch **65**, **66** and at least one left notch **67**, **68** on both sides of the circuit protection device **300**, by which stresses of thermal expansion are mitigated, and the device deformation is prevented such that the issue of poor resistance recovery can also be avoided. Furthermore, because the present invention uses packaging tape **90** or electrically insulating frame **91** to encapsulate the circuit protection device, environment factors can be blocked from contacting the circuit protection device and the entire structure of the circuit protection device may be further stabilized.

(36) The above-described embodiments of the present invention are intended to be illustrative only. Numerous alternative embodiments may be devised by persons skilled in the art without departing from the scope of the following claims.

Claims

1. A circuit protection device having an upper surface, a lower surface opposite to the upper surface, and a peripheral wall connected between the upper surface and the lower surface, the circuit protection device comprising: a first temperature sensitive resistor comprising a first upper

electrically conductive layer, a first lower electrically conductive layer, and a first positive temperature coefficient (PTC) layer laminated between the first upper electrically conductive layer and the first lower electrically conductive layer; a second temperature sensitive resistor comprising a second upper electrically conductive layer, a second lower electrically conductive layer, and a second PTC layer laminated between the second upper electrically conductive layer and the second lower electrically conductive layer; an electrically insulating multilayer having an upper insulating layer, a middle insulating layer, and a lower insulating layer, wherein: the upper insulating layer extends beyond the first upper electrically conductive layer, whereby the first upper electrically conductive layer is entirely covered and the first PTC layer is partially covered by the upper insulating layer; the middle insulating layer is laminated between the first lower electrically conductive layer and the second upper electrically conductive layer, thereby bonding the first temperature sensitive resistor to the second temperature sensitive resistor; and the lower insulating layer extends beyond the second lower electrically conductive layer, whereby the second lower electrically conductive layer is entirely covered and the second PTC layer is partially covered by the lower insulating layer; a right notch and a left notch, wherein the upper insulating layer is attached between the first electrode layer and the first upper electrically conductive layer, so as to space the first electrode layer apart from the first upper electrically conductive layer by a distance, and wherein the first electrode layer and the first upper electrically conductive layer are parallel to each other, and extend to the right notch, thereby electrically connecting to each other through the right notch; a first electrode layer and a second electrode layer attached to the upper insulating layer and the lower insulating layer, respectively, wherein the first electrode layer is electrically connected to the first upper electrically conductive layer and the second upper electrically conductive layer, and the second electrode layer is electrically connected to the first lower electrically conductive layer and the second lower electrically conductive layer; and a first external electrode disposed on the first electrode layer, wherein the first external electrode extends beyond the peripheral wall along a first horizontal direction parallel to the first electrode layer.

2. The circuit protection device of claim 1, wherein the distance ranges from 0.02 mm to 0.06 mm.
3. The circuit protection device of claim 1, wherein an opening is formed on the first electrode layer, and is positioned directly above where the first PTC layer is partially covered by the upper insulating layer, whereby the first electrode layer does not extend to the left notch and has no electrical connection to the left notch.
4. The circuit protection device of claim 3, further comprising an electrically insulating material, wherein the electrically insulating material is filled in the opening.
5. The circuit protection device of claim 3, wherein the right notch has a right electrically conductive member electrically connecting to the first electrode layer, the first upper electrically conductive layer, and the second upper electrically conductive layer, and wherein the left notch has a left electrically conductive member electrically connecting to the second electrode layer, the first lower electrically conductive layer, and the second lower electrically conductive layer.
6. The circuit protection device of claim 5, further comprising a structural reinforcing metal film, wherein the structural reinforcing metal film covers a surface of the first electrode layer, a surface of the second electrode layer, a surface of the right electrically conductive member, and a surface of the left electrically conductive member, whereby the first electrode layer, the second electrode layer, the right electrically conductive member, and the left electrically conductive member are insulated from environmental air.
7. The circuit protection device of claim 6, wherein the peripheral wall has a left sidewall, a right sidewall opposite to the left sidewall, a front sidewall, and a rear sidewall opposite to the front sidewall, wherein the left notch is disposed on the left sidewall, and the right notch is disposed on the right sidewall.
8. The circuit protection device of claim 7, wherein the first external electrode extends beyond the right sidewall along the first horizontal direction.

9. The circuit protection device of claim 8, wherein the circuit protection device has a first length parallel to the first horizontal direction, and the first external electrode has a second length from and beyond the right sidewall, wherein if the sum of the first length and the second length is calculated as 100%, the second length accounts for 31% to 53%.
10. The circuit protection device of claim 9, wherein the second length ranges from 4 mm to 8 mm.
11. The circuit protection device of claim 10, wherein the upper surface of the circuit protection device has a first upper surface area, an upper surface of the first external electrode has a second upper surface area, and a value by dividing the second upper surface area by the first upper surface area ranges from 1.37 to 1.64.
12. The circuit protection device of claim 8, further comprising a second external electrode disposed on the second electrode layer covered by the structural reinforcing metal film, wherein the first external electrode extends along a second horizontal direction parallel to the first electrode layer and beyond the front sidewall, and the second external electrode extends parallel to the first external electrode and beyond the rear sidewall.
13. The circuit protection device of claim 8, further comprising a second external electrode disposed on the second electrode layer covered by the structural reinforcing metal film, wherein the second external electrode extends opposite to the first horizontal direction and beyond the left sidewall.
14. The circuit protection device of claim 13, further comprising a packaging tape, wherein the packaging tape wraps around the circuit protection device so as to cover the upper surface, the front sidewall, the lower surface, and the rear sidewall, whereby only the right sidewall, the left sidewall, a part of the first external electrode, and a part of the second external electrode of the circuit protection device are exposed.
15. The circuit protection device of claim 13, further comprising an electrically insulating frame, wherein the electrically insulating frame encapsulates the circuit protection device, and only exposes a part of the first external electrode and a part of the second external electrode.
16. The circuit protection device of claim 1, further comprising a plurality of right notches and a plurality of left notches opposite to the plurality of right notches, wherein the plurality of right notches are disposed on a right sidewall and the plurality of left notches are disposed on a left sidewall of the peripheral wall, respectively, and the first external electrode extends beyond the right sidewall along the first horizontal direction.
17. The circuit protection device of claim 16, further comprising a second external electrode disposed on the second electrode layer, wherein the second external electrode extends opposite to the first horizontal direction parallel to the second electrode layer, and extends beyond the left sidewall.
18. The circuit protection device of claim 17, further comprising a packaging tape, wherein the packaging tape wraps around the circuit protection device, whereby only the right sidewall, the left sidewall, a part of the first external electrode, and a part of the second external electrode of the circuit protection device are exposed.
19. The circuit protection device of claim 17, further comprising an electrically insulating frame, wherein the electrically insulating frame encapsulates the circuit protection device, and only exposes a part of the first external electrode and a part of the second external electrode.
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